FM Radio — Dot Board Phase-1 "No Regrets" Build

RRD-102V2.0 (RDA5807M) + TDA2822M + Arduino UNO + 4-Ohm Speaker

Revision: 3.1 (added C7 10µF on 3.3V rail, 4.7k pull-up note, seek stability fix) **Date:** 2026-02-17

Design choices locked (Phase-0):

Decision	Choice
Power	Arduino USB 5V
I2C safety	External pull-ups to 3.3V (4.7k preferred; 10k OK)
Mono audio	Mix LOUT+ROUT via 2x 1k resistors
Amplifier mode	Single-ended Channel 1 (safest for dot board)
De-emphasis	50 us (Sri Lanka / Asia / Europe standard)
Tuning	Two push buttons: Seek Up + Seek Down (auto-find stations)
Power indicator	Red 5mm LED (always on when powered)

0) Absolute Rules — Read Before You Touch a Soldering Iron

1. **Never connect 5V to the RRD-102V2.0 VCC.** It is a 3.3V-only module. Connecting 5V will destroy the RDA5807M chip permanently.
2. **I2C pull-ups go to 3.3V only.** Do not pull SDA or SCL to 5V.
3. **Do not connect the speaker to the radio module.** The RDA5807M audio outputs are line-level (~360mV). They cannot drive a 4-ohm speaker. The TDA2822M amplifier must sit between the module and the speaker.
4. **Keep the antenna wire physically far from the speaker wires and amplifier section.** RF is sensitive; amplifier output current creates noise.

5. **Place decoupling capacitors close to each IC.** Within 1-2 cm. Distance matters in analogue circuits.
6. **Verify the TDA2822M orientation (notch/dot = pin 1) before soldering.** If the IC is backwards, it will not work and may be destroyed.
7. **LED polarity matters.** The longer leg is the anode (+). The shorter leg (or flat side of the body) is the cathode (-).

1) What You Are Building

1.1 Signal Path

```
FM Antenna (~75 cm wire)
        |
        v
  RRD-102V2.0  (RDA5807M FM tuner, 3.3V)
   LOUT + ROUT
        |
   R3 + R4 (1k + 1k mono mixer)
        |
       C1 (10uF coupling capacitor)
        |
        v
  TDA2822M  (audio amplifier, 5V)
   pin 3 (IN1+)  -->  pin 1 (OUT1)
        |
        v
   4-ohm Speaker
```

1.2 Control Path

```
BTN1 (Seek Up)   --> Arduino D2  (find next station upward)
BTN2 (Seek Down)  --> Arduino D3  (find next station downward)
Arduino A4 (SDA)  --> RDA5807M SDA  (I2C data)
Arduino A5 (SCL)  --> RDA5807M SCL  (I2C clock)
```

1.3 Power + Indicator

Rail	Source	Supplies
+5V	Arduino USB	TDA2822M, LED1 (power indicator)
+3V3	Arduino 3.3V pin	RRD-102V2.0, I2C pull-ups

LED1 (red, 5mm diffused) is always on when the system is powered. It is wired directly across the +5V rail through a current-limiting resistor (R6, 1k). No Arduino pin is used.

2) Parts List (BOM)

Everything below is the minimum for a reliable dot-board build. Tick off each part before you start.

2.1 Active Components

Ref	Part	Qty	Notes
—	Arduino UNO	1	Any compatible board
U1	RRD-102V2.0 (RDA5807M module)	1	Your module
U2	TDA2822M (DIP-8)	1	Use the 8-pin IC socket

2.2 Resistors

Ref	Value	Qty	Purpose
R1	4.7k (or 10k)	1	SDA pull-up to 3.3V (4.7k preferred)
R2	4.7k (or 10k)	1	SCL pull-up to 3.3V (4.7k preferred)
R3	1k	1	LOUT to mono mixer
R4	1k	1	ROUT to mono mixer
R5	2.2 ohm	1	Zobel network (output stability)
R6	1k	1	LED current limiting (gives ~3mA at 5V)

2.3 Capacitors

Ref	Value	Type	Qty	Purpose
C1	10uF 25V	Electrolytic	1	Audio coupling (mixer to amp)
C2	0.1uF (marked 104)	Ceramic	1	U1 VCC decoupling
C3	0.1uF (marked 104)	Ceramic	1	U2 VCC decoupling
C4	2200uF 50V	Electrolytic	1	5V bulk (near amplifier)
C5	0.1uF (marked 104)	Ceramic	1	Zobel network
C6	100uF 35V	Electrolytic	1	U2 pin 2 inverting-input bypass
C7	10uF 16V	Electrolytic	1	+3V3 rail bulk (RF stability, near U1)

2.4 Output + Indicator

Ref	Part	Qty	Notes
SPK1	4-ohm 5W speaker	1	
LED1	Red 5mm diffused LED	1	Power indicator (always on)

2.5 User Controls

Ref	Part	Qty	Notes
BTN1	Push button micro switch	1	Seek Up (next station)
BTN2	Push button micro switch	1	Seek Down (previous station)

2.6 Antenna

Part	Qty	Notes
Insulated wire ~75 cm	1	Quarter-wave for ~100 MHz centre

2.7 Mechanical / Misc

Part	Qty	Notes
8-pin DIP IC socket	1	Mount TDA2822M in socket (replaceable)
Dot board (perfboard)	1	Your 15x18 cm board
Hookup wire (solid core)	—	Multiple colours recommended

3) IC Pinouts — Get These Right or Nothing Works

3.1 TDA2822M (DIP-8) — Correct Pinout

TOP VIEW — notch/dot at pin 1 end:

```
        ____
OUT1  1|o   |8  VCC  (+5V)
IN1-  2|    |7  OUT2  (not used)
IN1+  3|    |6  IN2-  (tie to GND)
 GND  4|____|5  IN2+  (tie to GND)
```

Pin	Name	Function
1	OUT1	Audio output channel 1 --> speaker +
2	IN1-	Inverting input ch1 --> 100uF cap to GND
3	IN1+	Non-inverting input ch1 --> AUDIO IN (this is where your signal goes)
4	GND	Ground (only true GND pin)
5	IN2+	Non-inverting input ch2 --> tie to GND (unused)
6	IN2-	Inverting input ch2 --> tie to GND (unused)
7	OUT2	Output ch2 --> leave unconnected
8	VCC	Power supply --> +5V

WARNING: Many internet diagrams label pins 3/4/5 incorrectly. Trust the table above (source: ST Microelectronics TDA2822 datasheet, DIP-8 package).

3.2 RRD-102V2.0 Module Pad Order (Your Board)

Left side (top to bottom):

Pad	Label	Used?
1	SDA	YES
2	SCL	YES
3	NC	No
4	NC	No
5	VCC	YES (3.3V only!)

Right side (top to bottom):

Pad	Label	Used?
6	ANT	YES
7	NC	No
8	LOUT	YES
9	ROUT	YES
10	GND	YES

> **Stop and check:** If the silkscreen labels on your physical board differ from this table, follow your board's silkscreen. The pad order above is what you reported.

4) Complete Circuit — Every Connection

All connections are expressed as named nets so there is zero ambiguity.

4.1 Net Names

Net	Description
+5V	Arduino 5V pin (USB power)
+3V3	Arduino 3.3V pin
GND	Common system ground

Net	Description
SDA_3V3	I2C data line (pulled up to 3.3V)
SCL_3V3	I2C clock line (pulled up to 3.3V)
AF_L	Left audio output from radio module
AF_R	Right audio output from radio module
AF_MIX	Mono-mixed audio node (after R3+R4)
AF_IN	Amplifier input node (after coupling cap C1)
SPK+	Speaker positive terminal
SPK-	Speaker negative terminal (= GND)
BTN_UP	Seek-up button line (Arduino D2, pulled HIGH internally)
BTN_DN	Seek-down button line (Arduino D3, pulled HIGH internally)

4.2 Reference Designator Summary

Ref	Part	Value / Type
U1	RRD-102V2.0	RDA5807M FM module
U2	TDA2822M	DIP-8 audio amplifier
R1	Pull-up	4.7k preferred / 10k OK (SDA to +3V3)
R2	Pull-up	4.7k preferred / 10k OK (SCL to +3V3)
R3	Mixer	1k (AF_L to AF_MIX)
R4	Mixer	1k (AF_R to AF_MIX)
R5	Zobel	2.2 ohm
R6	LED limiter	1k (5V to LED1 anode)
C1	Coupling	10uF electrolytic
C2	Decoupling	0.1uF ceramic (U1)
C3	Decoupling	0.1uF ceramic (U2)
C4	Bulk	2200uF electrolytic

Ref	Part	Value / Type
C5	Zobel	0.1uF ceramic
C6	Bypass	100uF electrolytic (U2 pin 2)
C7	Bulk (3V3)	10uF electrolytic (near U1, +3V3 to GND)
LED1	Power indicator	Red 5mm diffused
BTN1	Seek Up	Push button micro switch
BTN2	Seek Down	Push button micro switch
SPK1	Speaker	4 ohm, 5W

4.3 U1 — RRD-102V2.0 (RDA5807M) Connections

Module Pad	Net / Destination
VCC	+3V3
GND	GND
SDA	SDA_3V3
SCL	SCL_3V3
LOUT	AF_L (to R3)
ROUT	AF_R (to R4)
ANT	75 cm antenna wire
NC pads	Leave unconnected

Place **C2 (0.1uF ceramic)** between +3V3 and GND, as close to U1 VCC/GND as possible. Place **C7 (10uF electrolytic)** between +3V3 (+) and GND (-), right next to C2. This bulk cap smooths USB switching ripple on the 3.3V rail, reducing I2C noise and random seek failures.

4.4 I2C Pull-Ups

Resistor	From	To
R1 (4.7k preferred, 10k OK)	SDA_3V3	+3V3

Resistor	From	To
R2 (4.7k preferred, 10k OK)	SCL_3V3	+3V3

Do NOT pull up to 5V. The RDA5807M is a 3.3V device. 5V on SDA/SCL will damage it.

> **Why 4.7k is better than 10k:** Faster rise times, cleaner SDA/SCL edges, and better noise immunity on perfboard with long hookup wires. If you already soldered 10k, it will work — no need to replace.

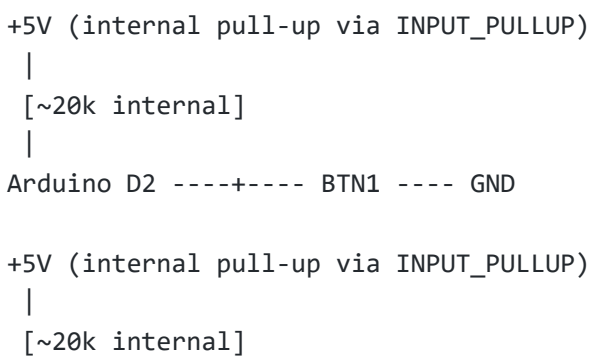
4.5 Arduino UNO Connections

Arduino Pin	Net / Destination
3.3V	+3V3 bus
5V	+5V bus
GND	GND bus
A4 (SDA)	SDA_3V3
A5 (SCL)	SCL_3V3
D2	BTN1 (Seek Up) — other leg to GND
D3	BTN2 (Seek Down) — other leg to GND

No external pull-up resistors are needed for D2/D3. The Arduino code enables internal pull-ups (~20k to 5V) on these pins.

4.6 Seek Buttons (BTN1 + BTN2)

Each button has two legs. One leg goes to an Arduino digital pin, the other leg goes to GND.



```
  |
Arduino D3 ----+---- BTN2 ---- GND
```

Button	One Leg	Other Leg	Arduino Pin	Action
BTN1	Arduino D2	GND	D2	Seek Up (find next higher station)
BTN2	Arduino D3	GND	D3	Seek Down (find next lower station)

When not pressed: pin reads HIGH (pulled up internally). When pressed: pin reads LOW (connected to GND through button).

4.7 Power LED (LED1)

The LED is a simple power indicator — always on when the system is powered. It does not use any Arduino pin.

```
+5V ---[ R6  1k ]---[>|--- GND
                    LED1
              (anode)  (cathode)
```

| Component | From | To | |-----------|------|----|| | R6 (1k) | +5V bus | LED1 anode (longer leg) | | LED1 cathode | LED1 cathode (shorter leg / flat side) | GND bus |

Current: (5V - 2V) / 1000 ohm = 3 mA. Dim but clearly visible on a diffused red LED. Safe, no heat.

4.8 Mono Mixer

Audio from the radio module is mixed to mono before entering the amplifier.

```
U1 LOUT ---[ R3  1k ]---+
                        |
                     AF_MIX ---[ C1  10uF ]--->  AF_IN
                        |
U1 ROUT ---[ R4  1k ]---+
```

Component	From	To
R3 (1k)	AF_L (U1 LOUT)	AF_MIX
R4 (1k)	AF_R (U1 ROUT)	AF_MIX

Component	From	To
C1 (10uF)	AF_MIX	AF_IN

C1 polarity: Positive (+) leg toward AF_MIX (radio side). Negative (-) leg toward AF_IN (amplifier side).

4.9 U2 — TDA2822M Amplifier Connections

Pin	Name	Connect To	Notes
1	OUT1	SPK+ (speaker positive)	Also connects to Zobel network
2	IN1-	C6 (100uF) to GND	Inverting input bypass for stable gain
3	IN1+	AF_IN	Audio signal input (from coupling cap C1)
4	GND	GND bus	The only ground pin on this IC
5	IN2+	GND	Unused channel — tie to ground
6	IN2-	GND	Unused channel — tie to ground
7	OUT2	Leave unconnected	Unused channel output
8	VCC	+5V bus	Power supply

C6 polarity: Positive (+) leg toward pin 2. Negative (-) leg toward GND.

4.10 Speaker

Terminal	Connect To
SPK+	U2 pin 1 (OUT1)
SPK-	GND

4.11 Zobel Network (Output Stability)

Prevents high-frequency oscillation on dot-board wiring.

```
U2 pin 1 (OUT1) ---[ R5  2.2 ohm ]---[ C5  0.1uF ]--- GND
```

R5 and C5 are in series, connected between OUT1 and GND.

4.12 Power Decoupling and Bulk Capacitors

Cap	Between	Placement
C2 (0.1uF ceramic)	+3V3 and GND	As close to U1 VCC/GND as possible
C7 (10uF electrolytic)	+3V3 (+) and GND (-)	Right next to C2, near U1. Reduces USB ripple on 3.3V rail
C3 (0.1uF ceramic)	+5V and GND	As close to U2 pin 8 and pin 4 as possible
C4 (2200uF electrolytic)	+5V and GND	Near U2, positive to +5V, negative to GND

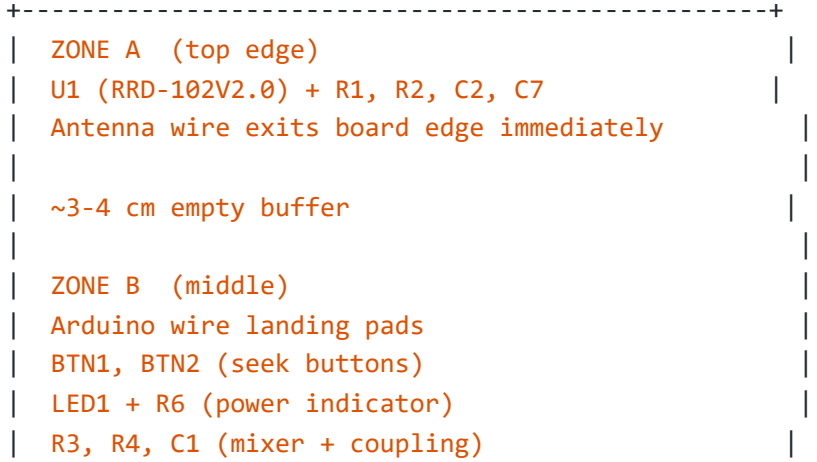
5) Dot-Board Layout Guidance

5.1 Board Sides

Side	What Goes There
Plain brown side	Components (ICs, caps, resistors, etc.)
Copper pad side	Solder joints and wire bridges

Components sit on the plain side. Their legs poke through the holes. You flip the board and solder on the copper side.

5.2 Physical Zoning (Top View of Component Side)



```
|                                                  |
|  ~2 cm buffer                                    |
|                                                  |
|  ZONE C  (bottom)                                |
|  U2 (TDA2822M in IC socket) + C3, C4, C6         |
|  R5, C5 (Zobel)                                  |
|  Speaker wires exit at bottom edge               |
+--------------------------------------------------+
|  LEFT EDGE  = +5V bus  (thick wire, vertical)    |
|  RIGHT EDGE = GND bus  (thick wire, vertical)    |
|  Short +3V3 bus near Zone A only                 |
+--------------------------------------------------+
```

Keep **at least 3-4 cm** between Zone A (radio) and Zone C (amplifier).

5.3 Ground Strategy

Create one **thick ground bus** (bare copper wire or thick hookup wire) running vertically along the right edge of the board. Bring ALL ground connections to this bus:

- U1 GND
- U2 pin 4 (GND)
- U2 pin 5, pin 6 (tied to GND)
- C2, C7, C3, C4, C5, C6 negative legs
- BTN1 one leg, BTN2 one leg
- Arduino GND
- Speaker negative

Do not daisy-chain grounds through thin wires from pad to pad. Every ground wire goes directly to the bus.

5.4 Wire Routing Rules

- **Antenna wire:** Exit the board edge immediately from U1 ANT pad. Do not route it across the board.
- **Speaker wires:** Keep short. Exit at bottom edge near U2. Do not route near Zone A.
- **Audio wires (AF_L, AF_R, AF_MIX, AF_IN):** Keep short. Route away from SDA/SCL.
- **SDA/SCL:** Lightly twist the pair together if practical. Keep away from audio wires.
- **Power wires (+5V, +3V3):** Run along the left edge bus. Keep away from audio/antenna.

6) Step-By-Step Build Procedure

Do not solder everything at once. Build in stages and verify at each stage.

Stage 1 — Lay Down Power Rails

1. Using thick wire on the copper side, create a **GND bus** along the right edge (vertical).
2. Create a **+5V bus** along the left edge (vertical).
3. Create a short **+3V3 bus** in Zone A only (a few pads near where U1 will sit).

Stage 2 — Install U1 (RRD-102V2.0) and I2C Pull-Ups

4. Mount and solder U1 in Zone A.
5. Solder **R1 (4.7k preferred, or 10k)**: from U1 SDA pad to +3V3 bus.
6. Solder **R2 (4.7k preferred, or 10k)**: from U1 SCL pad to +3V3 bus.
7. Solder **C2 (0.1uF ceramic)**: between +3V3 bus and GND bus, close to U1.
8. Solder **C7 (10uF electrolytic)**: positive (+) to +3V3 bus, negative (-) to GND bus, right next to C2.
9. Solder antenna wire to U1 ANT pad. Route the wire out of the board edge immediately.
10. Wire U1 VCC to +3V3 bus.
11. Wire U1 GND to GND bus.

Stage 3 — Install Mono Mixer, Buttons, and LED

12. Solder **R3 (1k)**: one end to U1 LOUT pad, other end to a shared node (AF_MIX).
13. Solder **R4 (1k)**: one end to U1 ROUT pad, other end to the same AF_MIX node.
14. Solder **C1 (10uF electrolytic)**: positive (+) to AF_MIX, negative (-) toward Zone C (this becomes AF_IN).
15. Mount **BTN1** (seek up) in Zone B. Wire one leg to a pad for Arduino D2, other leg to GND bus.
16. Mount **BTN2** (seek down) in Zone B near BTN1. Wire one leg to a pad for Arduino D3, other leg to GND bus.
17. Solder **R6 (1k)**: one end to +5V bus.
18. Solder **LED1** (red): anode (long leg) to the free end of R6, cathode (short leg / flat side) to GND bus.

Stage 4 — Install U2 (TDA2822M) and Support Components

19. Solder the **8-pin IC socket** in Zone C. Verify the notch orientation — pin 1 is at the notch end.

20. Solder **C3 (0.1uF ceramic)**: between +5V and GND, close to the socket (pins 8 and 4).
21. Solder **C4 (2200uF electrolytic)**: positive (+) to +5V, negative (-) to GND. Place close to socket.
22. Solder **C6 (100uF electrolytic)**: positive (+) to socket pin 2, negative (-) to GND.
23. Wire socket **pin 3 (IN1+)** to AF_IN (the negative end of C1 from step 14).
24. Wire socket **pin 4** to GND bus.
25. Wire socket **pin 5** to GND bus.
26. Wire socket **pin 6** to GND bus.
27. Wire socket **pin 8** to +5V bus.
28. Wire socket **pin 1 (OUT1)** to speaker positive pad.
29. Wire speaker negative pad to GND bus.
30. Solder **Zobel network**: R5 (2.2 ohm) from pin 1 to one leg of C5 (0.1uF), other leg of C5 to GND.
31. Leave socket **pin 7** unconnected.
32. **Do NOT insert the TDA2822M IC yet.** Leave the socket empty until all soldering near it is done.

Stage 5 — Wire to Arduino UNO

33. Run a wire from **+3V3 bus** to a pad labeled for Arduino 3.3V.
34. Run a wire from **+5V bus** to a pad labeled for Arduino 5V.
35. Run a wire from **GND bus** to a pad labeled for Arduino GND.
36. Run a wire from **U1 SDA pad** (same net as R1) to a pad labeled for Arduino A4.
37. Run a wire from **U1 SCL pad** (same net as R2) to a pad labeled for Arduino A5.
38. Run a wire from **BTN1** (the non-GND leg) to a pad labeled for Arduino D2.
39. Run a wire from **BTN2** (the non-GND leg) to a pad labeled for Arduino D3.
40. Connect these 7 pads to the Arduino using hookup wires with headers (or solder directly).

Stage 6 — Final Assembly

41. **Insert TDA2822M** into the IC socket. Pin 1 (notch end) must match the socket notch.
42. Inspect every solder joint on the copper side. Look for cold joints or bridges.
43. Verify LED1 is oriented correctly (long leg toward R6 / +5V side).
44. Proceed to pre-power checks (Section 7).

7) Pre-Power Checklist (MUST PASS Before

Applying Power)

Use a multimeter.

7.1 Continuity Checks (Power Off)

Test	Expected	If FAIL
+5V to GND	NOT shorted (no beep)	Find and fix solder bridge
+3V3 to GND	NOT shorted (no beep)	Find and fix solder bridge
SDA to GND	NOT shorted	Check R1 wiring
SCL to GND	NOT shorted	Check R2 wiring
U2 pin 1 to GND	NOT shorted	Check speaker/Zobel wiring
D2 pad to GND	NOT shorted (unless BTN1 pressed)	Check BTN1 wiring
D3 pad to GND	NOT shorted (unless BTN2 pressed)	Check BTN2 wiring

7.2 Voltage Checks (Power On via USB)

Connect Arduino to USB. Do NOT upload sketch yet.

Probe Point	Expected Voltage
U1 VCC pad	3.2V - 3.4V
U2 pin 8	4.8V - 5.2V
U2 pin 4	0V (ground)
LED1	Should be glowing red
BTN1 D2 pad (button not pressed)	~5V (internal pull-up)
BTN1 D2 pad (button pressed)	~0V

If U1 VCC shows ~5V, **disconnect immediately** — you have wired VCC to the wrong rail. If LED1 is not lit, check R6 orientation and LED polarity (anode toward R6).

7.3 Capacitor Polarity Verification (Visual)

Cap	Positive (+) Side	Negative (-) Side
C1 (10uF)	AF_MIX (toward radio module)	AF_IN (toward amplifier)
C7 (10uF)	+3V3 bus (near U1)	GND bus
C4 (2200uF)	+5V bus	GND bus
C6 (100uF)	U2 pin 2	GND bus

8) Arduino Code

Upload this sketch after passing all pre-power checks.

```
/*
 * FM Radio - RDA5807M via I2C + Seek Buttons
 * Phase-1 Dot Board Build (Rev 3.0)
 *
 * Wiring:
 *   Arduino A4 (SDA) --> RDA5807M SDA (with 4.7k/10k pull-up to 3.3V)
 *   Arduino A5 (SCL) --> RDA5807M SCL (with 4.7k/10k pull-up to 3.3V)
 *   Arduino D2       --> BTN1 (Seek Up)   other leg to GND
 *   Arduino D3       --> BTN2 (Seek Down) other leg to GND
 *   Arduino 3.3V     --> RDA5807M VCC
 *   Arduino 5V       --> TDA2822M pin 8, R6+LED1
 *   Arduino GND      --> common ground
 *
 * Usage:
 *   - Press BTN1 to seek to the next station (higher frequency)
 *   - Press BTN2 to seek to the previous station (lower frequency)
 *   - Open Serial Monitor at 9600 baud to see current frequency
 */

#include <Wire.h>

// ---- Pin definitions ----
#define BTN_SEEK_UP    2   // D2: seek next station (higher frequency)
#define BTN_SEEK_DOWN  3   // D3: seek previous station (lower frequency)

// ---- RDA5807M I2C address (sequential mode) ----
// Writes auto-start at register 0x02. Do NOT send register address byte.
static const uint8_t RDA_ADDR = 0x10;

// Shadow copies of registers 0x02 through 0x07
static uint16_t reg[8];
```

```
// ---- I2C Write: registers 0x02-0x07 ----
static void rdaWriteRegisters() {
  Wire.beginTransmission(RDA_ADDR);
  for (uint8_t r = 2; r <= 7; r++) {
    Wire.write((uint8_t)(reg[r] >> 8));   // high byte
    Wire.write((uint8_t)(reg[r] & 0xFF)); // low byte
  }
  Wire.endTransmission();
}

// ---- I2C Read: register 0x0A (2 bytes) ----
// In sequential read mode at 0x10, reads start at register 0x0A.
static uint16_t rdaReadReg0A() {
  Wire.requestFrom((uint8_t)RDA_ADDR, (uint8_t)2);
  uint8_t hi = Wire.read();
  uint8_t lo = Wire.read();
  return ((uint16_t)hi << 8) | lo;
}

// ---- Tune to a specific frequency (MHz) ----
static void rdaSetFrequency(float freqMHz) {
  if (freqMHz < 87.0f)  freqMHz = 87.0f;
  if (freqMHz > 108.0f) freqMHz = 108.0f;

  uint16_t channel = (uint16_t)((freqMHz - 87.0f) / 0.1f + 0.5f);

  // Register 0x03 layout:
  //   [15:6] CHAN, [5] DIRECT_MODE, [4] TUNE, [3:2] BAND, [1:0] SPACE
  reg[3] = ((channel & 0x03FF) << 6) | (1u << 4);

  rdaWriteRegisters();
  delay(100);

  reg[3] &= ~(1u << 4);  // clear TUNE bit
  rdaWriteRegisters();
}

// ---- Seek to next/previous station ----
// Returns the frequency (MHz) of the station found, or 0.0 if seek failed.
static float rdaSeek(bool seekUp) {
  // Set seek direction in reg 0x02
  if (seekUp) {
    reg[2] |= (1u << 9);   // SEEKUP = 1 (seek toward higher frequencies)
  } else {
    reg[2] &= ~(1u << 9);  // SEEKUP = 0 (seek toward lower frequencies)
  }
  reg[2] |= (1u << 8);     // SEEK = 1 (start seeking)
```

```
  rdaWriteRegisters();

  // Wait for STC (Seek/Tune Complete) flag in register 0x0A bit 14
  delay(200);
  uint16_t reg0A = 0;
  for (int attempt = 0; attempt < 80; attempt++) {
    reg0A = rdaReadReg0A();
    if (reg0A & (1u << 14)) break;  // STC = 1: seek/tune complete
    delay(50);
  }

  // Clear SEEK bit
  reg[2] &= ~(1u << 8);
  rdaWriteRegisters();

  // Dummy read to clear STC status latch (prevents rare seek locks)
  delay(50);
  rdaReadReg0A();

  // Check seek fail (bit 13 of reg 0x0A)
  if (reg0A & (1u << 13)) {
    // SF = 1: no station found (wrapped entire band)
    return 0.0f;
  }

  // Read the channel the chip stopped at
  uint16_t channel = reg0A & 0x03FF;
  float freqMHz = 87.0f + channel * 0.1f;

  // Update our shadow reg[3] to reflect the new channel
  reg[3] = (channel << 6);  // TUNE=0, BAND=00, SPACE=00

  return freqMHz;
}

// ---- Setup ----
void setup() {
  Serial.begin(9600);
  Wire.begin();
  delay(100);  // let RDA5807M power up

  // Configure button pins with internal pull-ups
  pinMode(BTN_SEEK_UP,   INPUT_PULLUP);
  pinMode(BTN_SEEK_DOWN, INPUT_PULLUP);

  // Register 0x02: DHIZ=1, DMUTE=1, ENABLE=1
  //   Bit 15: DHIZ     = 1  (audio output high impedance disabled = output ON)
  //   Bit 14: DMUTE    = 1  (not muted)
  //   Bit 13: MONO     = 0  (stereo; we mix externally with resistors)
```

```
  //   Bit 12: BASS     = 0
  //   Bits 6-4: CLK_MODE = 000  (32.768 kHz crystal, already on module)
  //   Bit 0:  ENABLE   = 1  (power up the chip)
  reg[2] = 0xC001;

  // Register 0x03: will be set by rdaSetFrequency()
  reg[3] = 0x0000;

  // Register 0x04:
  //   Bit 12: DE = 1         (50us de-emphasis, correct for Sri Lanka)
  //   Bit 10: SOFTMUTE_EN = 1
  reg[4] = 0x1400;

  // Register 0x05: volume + seek threshold
  //   Bits 3-0: VOLUME = 0xF (maximum volume = 15)
  //   Bits 11-8: SEEKTH = 0x8 (seek SNR threshold, default)
  //   Upper bits: common working defaults
  reg[5] = 0x88AF;

  reg[6] = 0x0000;
  reg[7] = 0x0000;

  rdaWriteRegisters();
  delay(200);

  // Tune to a default station (change 99.5 to a known local station)
  rdaSetFrequency(99.5);
  Serial.println(F("FM Radio Ready"));
  Serial.println(F("Tuned to 99.5 MHz"));
  Serial.println(F("Press BTN1 (D2) = Seek Up, BTN2 (D3) = Seek Down"));
}

// ---- Main Loop ----
void loop() {
  static bool lastBtnUp   = HIGH;
  static bool lastBtnDown = HIGH;

  bool btnUp   = digitalRead(BTN_SEEK_UP);
  bool btnDown = digitalRead(BTN_SEEK_DOWN);

  // Detect falling edge on Seek Up button
  if (btnUp == LOW && lastBtnUp == HIGH) {
    delay(50);  // debounce
    if (digitalRead(BTN_SEEK_UP) == LOW) {
      Serial.print(F("Seeking up... "));
      float freq = rdaSeek(true);
      if (freq > 0.0f) {
        Serial.print(freq, 1);
        Serial.println(F(" MHz"));
```

```
      } else {
        Serial.println(F("No station found"));
      }
    }
  }

  // Detect falling edge on Seek Down button
  if (btnDown == LOW && lastBtnDown == HIGH) {
    delay(50);  // debounce
    if (digitalRead(BTN_SEEK_DOWN) == LOW) {
      Serial.print(F("Seeking down... "));
      float freq = rdaSeek(false);
      if (freq > 0.0f) {
        Serial.print(freq, 1);
        Serial.println(F(" MHz"));
      } else {
        Serial.println(F("No station found"));
      }
    }
  }

  lastBtnUp   = btnUp;
  lastBtnDown = btnDown;

  delay(20);
}
```

8.1 If the Radio Does Not Respond

Problem A — No I2C communication at all: Some RDA5807M modules respond only to address 0x11 (index mode, which requires a register address byte). Change the address and write function:

```
static const uint8_t RDA_ADDR = 0x11;  // index mode

static void rdaWriteRegisters() {
  Wire.beginTransmission(RDA_ADDR);
  Wire.write((uint8_t)0x02);  // start register address
  for (uint8_t r = 2; r <= 7; r++) {
    Wire.write((uint8_t)(reg[r] >> 8));
    Wire.write((uint8_t)(reg[r] & 0xFF));
  }
  Wire.endTransmission();
}
```

If using 0x11 mode, also update `rdaReadReg0A()` to read from the correct register:

```
static uint16_t rdaReadReg0A() {
  Wire.beginTransmission(RDA_ADDR);
  Wire.write((uint8_t)0x0A);  // register address
  Wire.endTransmission(false);
  Wire.requestFrom((uint8_t)RDA_ADDR, (uint8_t)2);
  uint8_t hi = Wire.read();
  uint8_t lo = Wire.read();
  return ((uint16_t)hi << 8) | lo;
}
```

Problem B — Tunes but poor audio: Try enabling MONO in software by changing reg[2]:

```
reg[2] = 0xE001;  // bit 13 MONO=1
```

Problem C — Seek never finds stations:

- Confirm antenna is ~75 cm and connected to ANT pad.
- Lower the seek threshold (makes seek more sensitive to weak stations):

  ```
  reg[5] = 0x84AF;  // SEEKTH = 4 (lower = more sensitive)
  ```

- As a last resort, try tuning manually to a known local station:

  ```
  rdaSetFrequency(93.7);  // replace with your local station
  ```

9) Bring-Up Test Plan

Run these tests in order after uploading the code.

Test A — Power and LED

1. Plug in USB.
2. **LED1 should glow red.** If not, check LED polarity and R6 connection.
3. Measure U1 VCC: should read ~3.3V.
4. Measure U2 pin 8: should read ~5V.
5. Touch the TDA2822M IC briefly — it should be cool or barely warm.

Test B — Audio Chain Quick Test

1. With power on, lightly touch the AF_IN node with your finger.
2. You should hear a brief hum or pop from the speaker.
3. This confirms the amplifier is working.

Do not short AF_IN to power rails.

Test C — FM Reception (Seek Buttons)

1. Open the Serial Monitor at 9600 baud.
2. Press BTN1 (D2). The serial output should print "Seeking up..." then a frequency.
3. You should hear audio from the station found.
4. Press BTN2 (D3). It should seek down to the next lower station.
5. If seek wraps without finding anything, adjust the antenna or lower SEEKTH (see Section 8.1 Problem C).

Test D — Button Debounce

1. Quickly tap BTN1 several times. Each press should produce exactly one seek.
2. If you see multiple seeks per press, increase the debounce delay in the code (50 → 100 ms).

10) Troubleshooting

No sound at all

Check	Action
Speaker wiring	SPK+ to U2 pin 1, SPK- to GND
TDA2822M orientation	Notch/dot must be at pin 1
C1 installed?	Must be between AF_MIX and U2 pin 3
C6 installed?	100uF from U2 pin 2 to GND
U2 pin 4 to GND?	This is the only GND pin

Radio never tunes / no I2C

Check	Action
R1/R2 connected to 3.3V?	Must be +3V3, never +5V
SDA to A4, SCL to A5?	Swap if reversed
Try address 0x11	See Section 8.1 Problem A

Loud buzzing or digital ticking

Check	Action
Audio wire length	Shorten all audio wires
SDA/SCL near audio?	Re-route to separate them
C2 and C3 close to ICs?	Move closer if possible
Zobel installed?	R5+C5 from pin 1 to GND

Weak reception

Check	Action
Antenna connected?	Must be ~75 cm wire on ANT pad
Antenna position	Keep away from board, hold vertically
Try mono mode	Set reg[2] = 0xE001 in code

LED1 does not glow

Check	Action
LED polarity	Long leg (anode) toward R6 / +5V side
R6 installed?	1k between +5V bus and LED anode
GND connection	LED cathode (short leg / flat side) to GND bus

Buttons do not seek

Check	Action
Wiring	One leg to D2/D3, other leg to GND
Serial Monitor open?	9600 baud — look for "Seeking..." messages
Internal pull-up working?	Measure D2/D3 with button not pressed — should read ~5V
Button stuck low?	If pin always reads LOW, check for short to GND

TDA2822M gets very hot

Check	Action
Speaker impedance	Must be 4 ohm or higher
Short on output?	Check pin 1 is not shorted to GND
VCC correct?	Should be 5V, not higher

11) What Phase-1 Delivers

- Working FM radio reception (87-108 MHz)
- Mono speaker output through TDA2822M amplifier
- Two seek buttons: BTN1 (seek up) and BTN2 (seek down)
- Red LED power indicator (always on when USB connected)
- Serial Monitor frequency readout at 9600 baud
- Safe 3.3V I2C levels (no risk to RDA5807M)
- Stable USB-powered operation
- Replaceable amplifier IC (socketed)

Appendix A — One-Page Wiring Cheat Sheet

Print this page and keep it next to the dot board while soldering.

Arduino UNO Wires (7 wires to dot board)

Arduino Pin	Goes To
3.3V	+3V3 bus (powers U1, R1, R2, C7)
5V	+5V bus (powers U2, R6 → LED1)
GND	GND bus
A4	SDA_3V3 (U1 SDA pad)
A5	SCL_3V3 (U1 SCL pad)
D2	BTN1 one leg (other leg to GND)
D3	BTN2 one leg (other leg to GND)

RRD-102V2.0 (U1) Summary

Pad	Goes To
VCC	+3V3
GND	GND bus
SDA	Arduino A4 + R1 (4.7k/10k) to +3V3
SCL	Arduino A5 + R2 (4.7k/10k) to +3V3
LOUT	R3 (1k) to AF_MIX
ROUT	R4 (1k) to AF_MIX
ANT	75 cm wire antenna

TDA2822M (U2) Summary

Pin	Goes To
1 (OUT1)	Speaker + and Zobel (R5+C5 to GND)
2 (IN1-)	C6 (100uF) to GND
3 (IN1+)	AF_IN (from C1 negative leg)

Pin	Goes To
4 (GND)	GND bus
5 (IN2+)	GND
6 (IN2-)	GND
7 (OUT2)	Not connected
8 (VCC)	+5V bus

Passive Components at a Glance

Ref	Value	Between
R1	4.7k (or 10k)	SDA_3V3 --- +3V3
R2	4.7k (or 10k)	SCL_3V3 --- +3V3
R3	1k	U1 LOUT --- AF_MIX
R4	1k	U1 ROUT --- AF_MIX
R5	2.2 ohm	U2 pin 1 --- C5
R6	1k	+5V bus --- LED1 anode
C1	10uF	AF_MIX (+) --- AF_IN (-)
C2	0.1uF	+3V3 --- GND (near U1)
C3	0.1uF	+5V --- GND (near U2)
C4	2200uF	+5V (+) --- GND (-) (near U2)
C5	0.1uF	R5 --- GND
C6	100uF	U2 pin 2 (+) --- GND (-)
C7	10uF	+3V3 (+) --- GND (-), near U1
LED1	Red 5mm	Anode ← R6, Cathode → GND
BTN1	Micro SW	One leg → D2, other leg → GND
BTN2	Micro SW	One leg → D3, other leg → GND